

6 Old test from

6.2: Verify CloudFront Distribution Status

```
# Check distribution status
aws cloudfront get-distribution --id $CF_DIST_ID \
  --query "Distribution.Status"
```

```
# Check distribution status
aws cloudfront get-distribution --id E1TVVCTL7Q4QCW \
  --query "Distribution.Status"
```

Expected: `"Deployed"`

If status is `InProgress`: Wait 5-10 minutes and check again. CloudFront deployments can take 15-30 minutes.

6.3: Verify DNS Points to CloudFront (Not ALB)

```
# Check apex domain
dig williebright.com A +short

# Check app subdomain
dig app.williebright.com A +short
```

Expected: Returns CloudFront IPs (typically multiple IPs like `13.x.x.x`, `52.x.x.x`)

NOT Expected: ALB IPs (would show `us-east-1.elb.amazonaws.com` pattern)

```
# Verify CNAME chain points to CloudFront
dig williebright.com CNAME +short
```

Expected: Should show `*.cloudfront.net` domain

6.4: Verify Origin Cloaking (Direct ALB Access Blocked)

This is the **KEY TEST** for Lab 2A—direct ALB access must be blocked.

```
# Try to hit ALB directly (should FAIL with 403)
curl -I https://$ALB_DNS --insecure
```

Expected: HTTP/1.1 403 Forbidden

Response body should contain: Direct access forbidden

If you get 200 OK: Origin cloaking is NOT working. Check:

1. HTTPS listener default action is fixed-response 403
2. Listener rule with header condition exists
3. CloudFront is sending the secret header

6.5: Verify CloudFront Access Works

```
# Access via CloudFront (should SUCCEED)
curl -I https://williebright.com
curl -I https://app.williebright.com
```

Expected: HTTP/1.1 200 OK or HTTP/1.1 301/302 redirect to your app

If you get 403: Check WAF isn't blocking. If you get 502/504: Check ALB health and origin config.

6.6: Verify WAF is CloudFront-Scoped

```
# List CloudFront WAFs (must be in us-east-1)
aws wafv2 list-web-acls --scope CLOUDFRONT --region us-east-1 \
  --query "WebACLs[*].{Name:Name,Id:Id,Scope:Scope}"
```

Expected: Your WAF appears with scope CLOUDFRONT

```
# Get WAF details
aws wafv2 get-web-acl \
  --name helga-cf-waf01 \
  --scope CLOUDFRONT \
  --id $WAF_ID \
  --region us-east-1 \
  --query "WebACL.{Name:Name,Rules:Rules[*].Name}"
```

Expected: Shows WAF name and rules (RateLimitRule, AWSManagedRulesCommonRuleSet, etc.)

6.7: Verify CloudFront Has WAF Attached

```
aws cloudfront get-distribution --id $CF_DIST_ID \
  --query "Distribution.DistributionConfig.WebACLId"
```

Expected: Returns WAF ARN (not empty or null)

If empty: WAF is not attached to CloudFront. Check `web_acl_id` in your CloudFront distribution resource.

6.8: Verify ALB Security Group (CloudFront Prefix List Only)

```
# Get ALB Security Group ID
ALB_SG_ID=$(aws elbv2 describe-load-balancers \
  --query "LoadBalancers[?contains(LoadBalancerName, 'helg
a')].SecurityGroups[0]" \
  --output text)

# Check SG ingress rules
aws ec2 describe-security-groups --group-ids $ALB_SG_ID \
  --query "SecurityGroups[0].IpPermissions[*].{Port:FromPort,
PrefixListIds:PrefixListIds[*].PrefixListId,CidrIp:IpRanges
[*].CidrIp}"
```

Expected: Shows `PrefixListIds` with CloudFront prefix list (pl-xxxxxx), NOT `0.0.0.0/0`

If you see `0.0.0.0/0`: ALB is still open to the internet. Update the SG to use CloudFront prefix list only.

6.9: Verify Target Group Health

```
aws elbv2 describe-target-health --target-group-arn $TG_ARN
```

Expected: `State: "healthy"`

If unhealthy: EC2 instance may not be running or health check is failing. Check EC2 status and security groups.

6.10: Verify WAF Logging Active

```
# Check CloudWatch log group exists (us-east-1)
aws logs describe-log-groups \
  --log-group-name-prefix "aws-waf-logs-helga" \
  --region us-east-1 \
  --query "logGroups[*].{Name:logGroupName,Retention:retentionInDays}"
```

Expected: Log group exists with name starting with `aws-waf-logs-`

```
# Check WAF logging config
WAF_ARN=$(aws wafv2 list-web-acls --scope CLOUDFRONT --region
us-east-1 \
  --query "WebACLs[?contains(Name, 'helga')].ARN" \
  --output text)

aws wafv2 get-logging-configuration \
  --resource-arn $WAF_ARN \
  --region us-east-1
```

Expected: Returns logging configuration with CloudWatch log destination

6.11: Verify ACM Certificates

```
# CloudFront cert (us-east-1)
aws acm list-certificates --region us-east-1 \
  --query "CertificateSummaryList[?contains(DomainName, 'williebright')].{Domain:DomainName,Status:Status}"

# ALB cert (your deployment region)
aws acm list-certificates --region us-east-1 \
  --query "CertificateSummaryList[?contains(DomainName, 'williebright')].{Domain:DomainName,Status:Status}"
```

Expected: Status = `ISSUED` for both certificates

6.12: End-to-End Functional Test

```
# Full request through CloudFront → ALB → EC2
curl -v https://williebright.com 2>&1 | grep -E "HTTP/|Server:X-Cache"
```

Expected:

- `HTTP/2 200` or `HTTP/1.1 200`
 - `X-Cache: Miss from cloudfront` (first request) or `X-Cache: Hit from cloudfront` (cached)
 - `Server: CloudFront` header present
-